ADRENAL STEROIDS AND RELATED DRUGS

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Overview

- Adrenal glands are situated on top of the kidneys and are very essential for life

- They are composed of two major parts namely;

  1. Adrenal medulla
  2. Adrenal cortex

- The adrenal cortex is involved in the production of adrenal steroids or adrenocorticosteroids which are classified as ;

  - Mineralocorticoids
  - Glucorticoids
  - Adrenal androgens

- The adrenal medulla is involved in the production of epinephrine (Adrenaline) as its main hormone which is also a catecholamine, a group of chemical mediators very essential for life

Overview Cont'd

- In humans, **aldosterone** is the major mineralocorticoid, **cortisol** is the major glucorticoid while **dehydroepiandrosterone** (DHEA) is the major adrenal androgen

- The secretion of cortisol is regulated by corticotropin (ACTH) secreted by the pituitary gland

- On the other hand, aldosterone secretion is regulated by the renin angiotensin system

- Physical and mental stress are powerful activators of corticotropin releasing hormone secretion leading to increased corticotropin and cortisol production

- Cortisol exerts several effects that increase the body's resistance to stress

PHYSIOLOGICAL EFFECTS OF ADRENAL STEROIDS

- The adrenal steroids act on target tissues by binding to specific cytoplasmic steroid receptors
- They are then translocated to the cell nucleus, a common mechanism for most steroid hormones
- These actions lead to the various metabolic and anti-inflammatory effects of glucorticoid
- Renal tubular activation of mineralocorticoid receptor stimulates the synthesis of sodium channels and sodium potassium adenosine triphosphatase needed for sodium reabsorption
- Glucorticoids induce enzyme involved in gluconeogenesis and have an anti insulin effect thus explaining the hypoglycemic effect during stress in glucorticoid insufficiency
- Glucorticoids stimulate lipolysis and inhibit uptake of glucose by the adipose tissue, the mechanism that can lead to abnormal fat distribution and muscle wasting

Comparison of mineralocorticoid and glucorticoids receptors and their response to ligands

- The glucorticoid receptors have a high affinity for cortisol but a lower affinity for aldosterone
- Mineralocorticoid receptors has a higher affinity for both cortisol and aldosterone
- Thus cortisol exerts both glucorticoid and mineralocorticoid actions

Glucorticoids

- Most of the glucorticoids are readily available as; Oral, parenteral, inhalational or topical administration

- Topical or inhalational preps have been preferred in specific conditions as they are highly tolerable and tends to avoid most systemic adverse effects

Classification of glucorticoids

Glucorticoids are mostly classified based on ;

- Potency of the glucorticoids

- Duration of action of systemically administered drugs is mostly based on their potency at the receptor site where highly potent drugs tend to evoke a longer acting stimulation of gene transcription than the lesser potent

Low Potency – Immediate acting steroid

- **Cortisol** is the major glucorticoids in the body which is rapidly metabolized to hydrocortisone by the liver

- Pharmaceutically exists as **hydrocortisone**

- Cortisol has a half life of 8 – 12 hours and has equal glucorticoid and mineralocorticoid effects

- Very **useful** in replacement therapy in patients with adrenal insufficiency

Medium potency – Intermediate acting glucorticoids

- Examples - **prednisone**, **prednisolone**, **methylprednisolone** and **triamcinolone**
- In the body, prednisone is rapidly converted to prednisolone, a substance which itself exists also exists as a readily pharmaceutical drug
- These have half life of 12 to 36 hours and are used in cancer, inflammation, allergy and auto immune diseases

High potency – Long acting glucorticoids

- Examples are **betamethasone** and **dexamethasone** are stereoisomers differing only in the configuration of a methyl group
- Betamethsone is available for systemic use and topical treatment of a number of conditions like psoriasis, seborrheic or contact dermatitis and neurodermatitis
- Dexamethasone is used in dexamethasone suppression test and a mangemnt of a number of neoplastic, infectious and other inflammatory conditions requiring potent and long acting drugs
- **Budenoside** is a long acting acting glucorticoid which is administered through the inhalational route

INDICATIONS

- Glucorticoids are used for the treatment and diagnosis of adrenal diseases and also treatment of none adrenal conditions

Inflammation , Allergy and Autoimmune disorders

- Suppresses inflammation and immune dysfunction associated with diseases affecting almost every organ in the body

- Counteracts inflammation evoked by trauma, extreme temperatures, noxious chemicals, radiation damage and microbial pathogens

- Examples of diseases treated with corticosteroids are lupus erythematosus, autoimmune thrombocytopaenia purpura, poly arteritis nodosa, multiple sclerosis, ulcerative colitis, polymyositis

Inflammation , Allergy and Autoimmune disorders (Cont'd)

- Beclometasone and mometasone are available for nasal insufflation in the treatment of asthma and allergic rhinitis respectively
- For corneal keratitis and other pain associated to ocular surgery, many glucorticoids are available in eye drops and mostly combine with antimicrobial agents like neomycin ,gentamycin or tobramycin
- Similarly, other formulations exist as ear drops for superficial bacterial infections of the external auditory canal

Cancer

- Glucorticoids are used in lymphocytic leukamias and lymphomas because of their lymphotoxic effects

- Dexamethasone is used in combination with other drugs to prevent emesis during the administration of chemotherapy (Pre-medication drug)

Respiratory distress syndrome

- Betamethsome is used to prevent respiratory distress syndrome in premature infants

- It enhances fetal lung maturation just like endogenous cortisol does

- Beclometasone is preferred because it is not highly protein bound and hence can easily cross the blood brain barrier

Adrenal insufficiency

Primary adrenal insufficiency (addisons disease)

- Has all regions of the adrenal cortex destroyed
- This gives a deficiency in cortisol and aldosterone and to a reduction in androgen secretion

Secondary adrenal

- This type of insufficiency has several causes, the prolonged usage of steroid drugs being one of the notable ones

Mechanism by which steroid usage causes adrenal insufficiency

- Prolonged use of steroid drugs usually suppresses the hypothalamic pituitary adrenal axis where there is impaired inhibitory feedback for corticotropin releasing hormone resulting in inhibition of production of endogeneous cortisol

- This is characterized by lower levels of cortisol and androgens but normal levels of aldosterone

Acute adrenal insufficiency (Adrenal crisis, addisonian crisis)

- A medical emergency whose treatment must be prompt with intravenous hydrocortisone for up to 48hours

- After stabilization of patients condition, long term oral hydrocortisone can be commenced

- Hydrocortisone is administered orally in a manner that mimics the circadian secretion of cortisol by the adrenal glands with two thirds given in the morning and one third in the evening

Anti-inflammatory effects of corticosteroids

- There are a number of mechanism by which glucorticoids effect their role in of anti-inflammation which sometimes are interlinked;

ONE

- Prevention of activation of lymphocytes by interleukins and nuclear factors, a transcription factor for pro-inflammatory cytokine genes

TWO

- Suppression of cytokines by activated helper T cells and thus preventing recruiting and activation of eosinophil and stimulation of anti body production by B cells

THREE

- Decrease of the release of various chemical mediators of inflammation, including histamine, prostaglandins, leukotrienes and other substances from mast cells, oesinophils and inflamed tissue

- Decrease the synthesis of prostaglandin and leukotrienes by inhibiting the transcription of annexin 1(lipocortin 1) which inhibits phospholipase A, a first step in the eicosanoid pathway

Anti-inflammatory effects of corticosteroids Cont'd

FOUR

- Stabilization of lysosomal membranes of neutrophils and prevent the release of catabolic enzymes like phosphatase and hence limiting cytotoxic effects of inflammation

FIVE

- Glucorticoids cause vasoconstriction and decrease capillary permeability by increased annexin 1 leading to decreased prostacyclin

SIX

- Glucorticoids decrease activation of macrophages and decrease the synthesis and release of cytokines from these cells

- Furthermore, glucorticoids also affects the circulating leukocytes and hence the pharmacologic doses suppressing the lymphoid tissue and reduce the number of circulating lymphocytes

Congenital adrenal hyperplasia(CAH)

- A group of disorders caused by specific enzyme deficiencies that impair the synthesis of cortisol and aldosterone

- This leads to compensatory increase in corticotropin secretion by the pituitary

- Because of the deficiency of enzymes, the steroid biosynthesis pathway shifts to the production of androgens

- Androgens lead to virilization (Masculinization) and pseudohermaphroditism in girls and precocious development of secondary sex charecteristics including genitals (macrogenitosomia) in boys

- Treament of CAH is done using hydrocortisone which works by suppressing the secretion of corticotropin

- Fludrocortisobne can also be given for salt losing patients with CAH

Cushing syndrome

- This is a condition that involves the adrenocortical hyperfunction ,mostly caused by excessive circulating levels of corticotropin
- Pituitary adenomas produce excessive quantities of corticoptropin and hence leading to adrenal hyperplasia and excessive cortisol production
- Adrenal adenomas, adrenal carcinomas and ectopic corticotropin (ACTH) secreting tumors are also known to causes of cushing syndrome

Symptoms of Cushing syndrome

- **Moon face** (Rounded puffy face) due to redistribution of fat fromfrom the extremities to the face and trunk
- **Buffalo hump** – fat accumulation in the supraventricular and dorsocervical areas
- Increased hair growth (Hirsutism)
- Weight gain
- Muscle wasting and weakness

Diagnosis of Cushing syndrome

- Free cortisol level in urine
- Dexamethasone suppression test

In low dose dexamethasone suppression test

- A single dose is given and after 7 to 8hrs then a test is done
- In individuals without cushing syndrome, there will be a suppreesion effect on the pituitary cortisol secretion of up to 5mcg/dL
- In patients with cushing syndrome, corticotropin secretion is not suppressed but rather exceed up to 10mcg/dL suppression

Treatment

- Surgical excision of the pituitary adenoma or the hyperplastic adrenal glands
- High dose parenteral administration during the surgery then the dose is tapered to a normal replacement level

Dermatologic conditions

- Corticosteroids are used to treat a number of dermatologic conditions like atopic (contact) and seborrheic dermatitis, pruritus (itching) of different causes, psoriasis, sun burn

- Topical corticosteroids are grouped according to their relative anti-inflammatory potency i.e

1. High potency
2. Medium potency
3. Low potency

Low potency drugs

- Examples – Hydrocortisone, desonide, dexamethasone
- Preferred for treating areas with thinner skin e.g skin ,eyes and the intertriginous areas where skin is folded or overlapped
- Allergic reactions like insect bites are one of the specific examples of conditions treated

Low to medium potency steroids

- Examples – triamcinolone, fluticasone
- can be used on the ears,trunk, arms, legs and scalp

Medium to very high potency steroids

- Examples – desoximetasone and fluocinonide (High potency) while betamethasone dipropionate are very high potency steroids
- drugs may be used to treat areas of thicker skin like the palms and soles

Other specific disorders and indications

- Glucorticoids are used to treat hypercalcemia and sarcoidosis (systemic granulomatous disorder)
- Glucorticoids are used as immunosuppressant drugs to prevent organ graft rejection
- Allergic reactions
- Autoimmune diseases e.g systemic lupus erythematosus
- Neoplastic diseases

Systemic administration and pharmacokinetics

- Glucorticoids are highly lipid soluble and well absorbed from the gut after oral administration
- In circulation, glucorticoids are highly bound to corticosteroids binding globulin and albumin
- Glucorticoids are administered orally for treatment of allergic reactions, autoimmune disorders, neoplastic diseases and other conditions
- For acute disorders, glucorticoids are more effective when initially given in higher doses which are gradually tapered until treatment is discontinued
- For severe autoimmune and inflammatory disorders like systemic lupus erythematosus, large doses of prednisolone must be given daily for several months until a remission is achieved then the dose is slowly tapered and continued up to 1 to 2 years
- In some cases are switched to **alternate – day therapy** where all or most of the doses are administered on alternative days and this appears to reduce severity of adverse effects and produces less suppression of the hypothalamic pituitary adrenal axis by allowing more time for recovery in between the doses

Adverse effects of corticosteroids

- Usage of supraphysiological doses of glucorticoids for more than 2 weeks produces a series of tissue and metabolic changes that result in a form of Cushings like syndrome

- Other metabolic and physiological changes caused by glucorticoids include hyperglycemia, glucose intolerance, hypertension

- Sodium retention, potassium loss and hypertension are more pronounced when cortisone and hydrocortisone are used as they have more mineralocorticoid activity than other glucoiticoids

Adverse effects of corticosteroids Cont'd

- Glucorticoids increase bone catabolism and interferes with the effect of vitamin D on calcium absorption and hence the pronounce osteoporotic effect among long term users

- On the CNS, there is alteration of moods in some individuals and can either lead to euphoria or psychosis

- Peptic ulcer disease – due to stimulated gastric acid and pepsin production caused by large doses of glucorticoids

- Long term use of glucorticoids can lead to Posterior subcapsular cataract and glaucoma

- Can mask symptoms and signs of mycosis and other infections

NB:

- Glucorticoids should therefore be avoided or used with caution in patients with psychosis, heart disease, hypertension, diabetes, PUD, osteoporosis and certain infections

ANALGESIC DRUGS

INTRODUCTION

- Pain is an unpleasant sensory and emotional experience associated with the actual or potential tissue damage or described in terms of such damage
- Pain can be classified as acute or chronic
- Pain is further, described as mild, moderate and severe

ANALGESICS

- These are drugs which relieve pain caused by multiple causes

- They are classified into two ;

i. Non opioid analgesics used in mild to moderate pain e.g. Non Steroidal Anti-inflammatory drugs(NSAIDs) and paracetamol

ii. Opioid analgesics which are mostly used in moderate to severe pain e.g. morphine , synthetic opioids

iii. Adjuvants: these are medication that are not primarily designed to control pain but can be used for this purpose of pain but can be used for this purpose and are thus prescribed in addition to other medications or on its own

Examples of adjuvants are anticonvulsants like carbazepam, gabapentin, pregabalin and antidepressants like amitriptyline, imipramine

Differences among NSAIDs

- Differences in anti inflammatory activity between NSAIDs are small but there is considerable variation in individual patient tolerance

- The main difference is in the incidence and type of side effects

Difference between NSAIDs and Opioids

- NSAIDS have a ceiling effect in that more analgesic effects is exerted with the increase in dose but will reach a certain dose where no more improved effect can be attained

- On the other end, the more the dose of opioids is increased the higher the effect they give of course with consequent increase in side effects but has no ceiling effect

NSAIDs AND PARACETAMOL

- NSAIDs (Non steroidal anti-inflammatory drugs) are a group of agents that share the common capacity to induce Anti-inflammatory, analgesic and anti-pyretic effects
- On the other hand, paracetamol only has analgesic and anti pyretic effect with no anti-inflammatory action

Mechanisms of action

- Acetyl salicylic acid(Aspirin) induces irreversible inhibition of both cyclo-oxygynase enzyme (Both COX - 1 and COX – 2)
- Other NSAIDS cause competitive reversible inhibition of COX enzymes
- **Paracetamol inhibits prostaglandin (PG) synthesis in CNS resulting in Analgesic and Antipyretic** effects but is a weak inhibitor of PG in periphery and thus, it has no Anti-inflammatory effect

Acetyl Salicylic acid (Aspirin)

Pharmacological actions and Therapeutic uses of Aspirin:

1- **Anti-inflammatory**: Decreases vasodilator PGE2 and PGI2 thus reducing the vasodilation and edema of inflammation

- Anti-oxidant effect (free radical scavenger)

Uses: - Rheumatic fever, Rheumatoid arthritis

2- **Analgesic effect:**

Peripheral effect: Through its anti-inflammatory effect, it decreases PGs which sensitize nerve endings to Kinins (released during inflammation) thus increasing pain threshold.

Central effect: Decreases PGs thus inhibiting pain stimuli at sub-cortical sites.

Uses: - Mild to moderate pain especially secondary to inflammation; Arthritis, Dental pain, Headache (decreases cerebral vasodilator effect of PGs), Dysmenorrhea and Postpartum pain

3- **Antipyretic effect:**

Decreases PGE2, which is generated in response to inflammatory pyrogens (e.g. Interleukin-1) and which is responsible for elevating the hypothalamic set point for temperature control

Uses: - Antipyretic (Lowers body temperature) in fever

4- **Antiplatelet effect:**

Irreversible inhibition of COX enzyme in platelets results in inhibition of TXA2 (Thrombohexane A2) synthesis and inhibition of platelet aggregation.

Uses: - Prophylaxis against Transient Ischemic Attacks, Myocardial infarction and unstable Angina.

Adverse effects:

- **Gastro- intestinal side effects**: - Dyspepsia, nausea, vomiting, gastritis, ulceration with risk of hemorrhage (due to direct irritant effect on the mucosa and decreased protective PGs).

- **Renal damage (Analgesic nephropathy)**: due to decreased renal vasodilator PGs.

- **Hypersensitivity reactions**: Skin rash, Rhinitis, Asthma especially in asthmatics and patients with nasal polyps (*Aspirin diverts arachidonate metabolism from* cyclo-oxygenase to lipoxygenase pathway increasing chemotactic LTB4 and spasmogenic LTs).

- **Hyper-uricemia in patients with Gout (in low doses):** Competes with uric acid for excretion by organic acid secretory mechanism in renal tubules.

- **Reye's syndrome** (Encephalopathy and liver damage) in children with fever due to viral infection
- **Increased bleeding tendency**: caused by the anti-platelet effect
- **Chronic toxicity (Salicylism):** Large doses used for long periods lead to dizziness, tinnitus and gastric upset.
- **Acute toxicity:** Respiratory alkalosis (hyperpnoea and washing out of CO_2) followed by respiratory acidosis due to respiratory depression and metabolic acidosis (due to accumulation of acidic metabolites).

Other NSAIDs

Ibuprofen

- Effective and better tolerated (decreases incidence of side effects than other NSAIDs)
- choice in inflammatory joint disease

Naprofen

- It is related to *Ibuprofen, more potent with moderate risk of side effects.*
- It is longer acting (given twice daily)

Piroxicam

- Potent and long acting ($t1/2$ = 45 hours); given once daily
- No accumulation in the elderly or in patients with renal impairment
- 20% of patients develop side effects e.g. GIT bleeding

Diclofenac

It is very potent and used in:

- Long-term treatment of chronic inflammatory musculoskeletal disorders e.g. Rheumatoid arthritis, Osteoarthritis and Ankylosing Spondylitis
- Renal colic and Post-operative pain
- An ophthalmic solution is used for post-operative inflammation

Indomethacin

Potent, but due to its serious adverse effects, its use is limited to:

1- Acute Gouty arthritis

2- Rheumatoid arthritis, Ankylosing Spondylitis

3- Post-operative pain.

4- Patent ductus arteriosus (inhibits PG synthesis closing the ductus)

Adverse effects of Indomethacin:

1) GIT: Nausea, vomiting, ulcer, bleeding

2) CNS: Dizziness, confusion, ataxia, severe headache (cerebral vasodilatation)

3) Salt and water retention (antagonize anti-hypertensives) and hyperkalemia. It aggravates pre-existing renal failure.

- 4) Aplastic anemia

SELECTIVE COX-2 INHIBITORS:

Colecoxib and Rofecoxib

- **These are selective COX-2 inhibitors that spare COX-1, thus** they do not inhibit synthesis of protective PGs in the GIT and hence having less GIT side effects
- **Colecoxib is structurally related to Sulphonamides and therefore, its use might be** associated with development of skin rash

PARACETAMOL (ACETAMINOPHEN)

- Paracetamol has analgesic and antipyretic effect with no anti-inflammatory action.

- Preferred to *Aspirin in:*
  1. Patients allergic to *Aspirin*
  2. Peptic ulcer (No GIT disturbances)
  3. Gout (Aspirin may cause hyper-uricemia)
  4. Viral infection in children (to avoid Reye's syndrome with *Aspirin)*
  5. Bleeding disorders (does not affect platelet function)

- Paracetamol is ineffective in high fever because excess free radicals generated by leucocytes antagonize Antipyretic effect of Paracetamol.

- **Adult Dose: - It is given orally (500mg 2-3 times daily) and is metabolized in the liver**

PARACETAMOL HEPATOTOXICITY

- Toxic doses (15 grams or more) cause nausea and vomiting, followed in 24-48hrs by potentially fatal liver damage due to generation of a toxic metabolite.
- In toxic doses, saturation of normal conjugation enzymes results in conversion of the drug by mixed function oxidases to a toxic metabolite (N-acetyl-p benzoquinone).
- If this toxic metabolite is not inactivated by conjugation with Glutathione, it reacts with cell proteins and kills the cell.

Treatment:

- Agents which increase Glutathione (N-acetylcysteine orally or I.V).
- Agents which increase conjugation reaction (Methionine) *can prevent* liver damage if given early.

END